

When Is Label-Free Adaptation Knowable? Helpful, Harmful, and Unknowable Regimes Under Distribution Shift

Anonymous Authors

Paper under double-blind review

Abstract—Test-time adaptation can improve models on unlabeled target data, but the same objective that helps under one distribution shift can silently hurt under another. Existing work mainly strengthens adaptation mechanisms; we study a prior decision problem: can a system decide, without target labels, whether it should adapt at all?

We formalize adaptation benefit as the excess target risk of a frozen baseline over an adapted candidate and define three label-free regimes: *knowably helpful*, *knowably harmful*, and *unknowable*. We prove that when two target worlds induce identical observable evidence but opposite adaptation benefit, no label-free rule can be correct in both; abstention is then information-theoretically necessary. Under a risk-alignment assumption, we give a finite-sample adapt/freeze/abstain certificate controlling false adaptation and false freezing at level α under stated calibration assumptions, and an exact benefit-sign frontier: adaptation benefit is identifiable from observables iff an observable margin exceeds the calibration-drift budget on the disagreement region.

We instantiate the framework as KGA, a mechanism-agnostic wrapper around Tent, EATA, and SAR, and evaluate it on a *six-dataset core panel* (Office-Home, Camelyon17, CIFAR-10-C, iWildCam, CIFAR-10.1, ImageNet-R). KGA beats both always-adapt and always-freeze exactly in harmful-detectable regimes—a *statistically robust* beats-both, with bootstrap 95% confidence intervals that exclude zero on Office-Home, Camelyon17, and the CIFAR-10-C stress grid—while *matching or remaining close to* the better trivial policy in benign or weak-evidence regimes (Table I). On a heterogeneous mixed-deployment stream pooling the held-out shifts, its regret is 14–27 $\times$ below either fixed policy. KGA is not a universal accuracy booster; reported wins are regime-specific.

Index Terms—test-time adaptation, distribution shift, selective prediction, identifiability, label-free risk estimation

I. INTRODUCTION

Deployed models rarely encounter the same distribution they were trained on. Test-time adaptation (TTA) updates a model on unlabeled target data, often through entropy minimization or stability-aware updates. In favorable shifts, adaptation can recover accuracy; in unfavorable shifts the same objective can silently degrade performance. Because

target labels are unavailable, the system often cannot tell whether adaptation is helping or hurting.

Most work asks how to design a better adapter. We ask whether to adapt at all. Given a frozen baseline f_0 , an adapted candidate f_a , and an unlabeled batch, the system chooses **adapt** when adaptation is knowably helpful, **freeze** when knowably harmful, or **abstain** when label-free evidence cannot identify the sign of the benefit.

Identifiability constrains this decision. There exist target worlds with identical label-free evidence but opposite adaptation benefit; then no label-free rule can be correct in both worlds and abstention is mandatory. When evidence is risk-aligned and a benefit estimate carries a valid uncertainty radius, a finite-sample certificate can route between adapt, freeze, and abstain while controlling false-adapt and false-freeze errors at level α under exchangeable calibration.

We instantiate **one** method, KGA (Algorithm 1): a gradient-boosted benefit estimator $\hat{\Delta}(Z)$ plus a conformal radius ε fit on a labeled calibration split, then adapt/freeze/abstain. The wrapper is mechanism-agnostic (Tent, EATA, SAR); there is no separate multicandidate or alternate certificate in the headline experiments. K-Bound does *not* claim universal improvement: when adaptation is already safe, a certificate can only tie always-adapt; its value is highest when harmful adaptation is frequent, detectable, and costly.

a) Contributions.: We formalize label-free TTA as an adapt/freeze/abstain decision; prove non-identifiability of the benefit sign and a finite-sample certificate with an exact benefit-sign frontier; instantiate KGA as a wrapper around Tent/EATA/SAR; and validate KGA on a *six-dataset core panel*—Office-Home, Camelyon17, CIFAR-10-C, iWildCam, CIFAR-10.1, and ImageNet-R—with bootstrap confidence intervals on the held-out wins and a heterogeneous mixed-deployment capstone (Table I); ImageNet-C provides ImageNet-scale corroboration of the harmful-detectable regime.

Empirically, KGA follows the regime map: synthetic stress grids supply the main policy wins (CIFAR-10-C, ImageNet-C SAR); Camelyon17 Protocol G, iWildCam Protocol H v2, and OfficeHome Protocol M v2 are held-out real-shift beats-both replications under canonical dev-lock

TABLE I

Reviewer-safe dataset/regime evidence summary. NUMBERS ARE REGRET-TO-ORACLE (LOWER IS BETTER). HEURISTIC/AETTA/ATC/AGREEMENT COLUMNS ARE MARKED “NOT UNIFORM” WHEN NO ARTIFACT EXISTS UNDER THE SAME ADAPTER, SPLIT, AND CALIBRATION PROTOCOL; WE DO NOT IMPUTE BY ANALOGY. DIRECT COMMITTAL GATE BASELINES ON CONTROLLED TRACKS (ATC-STYLE CONFIDENCE ON IMAGENET-C SAR) ARE IN APPENDIX K, TABLE XVIII. NATURAL-SHIFT HEADLINE ROWS USE PROTOCOLS G / H v2 / M v2. THE DEPLOYED OFFICE-HOME ADAPTER AUDIT REMAINS A SEPARATE NULL; IMAGENET-R IS SPLIT-SPECIFIC ONLY.

Dataset/regime	always-freeze	always-adapt	heuristic gate	AETTA/ATC/agreement	KGA	oracle	false-adapt
CIFAR-10-C stress (Tent/EATA)	0.1202/0.1311	0.0086/0.0037	not uniform	not uniform	0.0016/0.0015	0	$0 \leq \alpha$
ImageNet-C SAR	0.0267	0.1124	not claimed	ATC worse; FA 1.00	0.0229	0	$0 \leq \alpha$
Camelion/T Protocol G	0.07492	0.00132	not uniform	not uniform	0.00006	0	$0.0256 \leq \alpha$
iWildCam Protocol H v2	0.00410	0.10283	not uniform	not uniform	0.00367	0	$0 \leq \alpha$
OfficeHome Protocol M v2	0.01582	0.04681	not uniform	not uniform	0.00220	0	$0 \leq \alpha$
ImageNet-R (lead only)	0.01802	0.00521	not uniform	not uniform	0.00417 (1/4 splits)	0	$0.0071 \leq \alpha$

KGA; elsewhere weak evidence correctly forces abstention or conservative freeze (Appendix).

Table note. The table does not impute heuristic/AETTA/ATC/agreement results where the corresponding artifact is not implemented under the same adapter, split, and calibration protocol; direct committal gate baselines on controlled tracks are reported separately (Appendix K).

II. RELATED WORK

a) *TTA methods.*: Tent [1], SHOT [2], TTT [3], EATA [4], SAR [5], CoTTA [6], and MEMO [7] improve robustness of adaptation but commit to adapting; none certifies the sign of benefit *before* updating or abstains on the adapt/freeze decision itself.

b) *Guarded and monitored TTA.*: POEM [11] and sequential risk monitors [9], [10] act during or after adaptation and control false alarms in one failure direction. K-Bound decides *before* committing, poses a three-way decision with an explicit unknowable region, and controls false-adapt on the benefit sign.

c) *Label-free accuracy and error estimation.*: ATC [13], Agreement-on-the-Line [15], [14], and AETTA [17] estimate target performance without labels. Most directly, [16] applies agreement-on-the-line to decide, at the *population* level, when TTA helps or hurts and to select an adaptation strategy; it issues no per-deployment certificate, no ABSTAIN action, and no identifiability limit, and its linear trend is known to break on shifts (e.g. Gaussian-noise corruption)—precisely the regime where K-BOUND abstains. Estimating $R_T(f)$ differs from certifying $\text{sign}(R_T(f_0) - R_T(f_a))$ before adapting with controlled false-adapt rate. Thus, K-Bound is not an accuracy estimator used as a gate; it changes the target estimand from target accuracy to the sign of the frozen-versus-adapted risk difference, with abstention when that sign is not identifiable.

d) *Identifiability, conformal inference, and selective prediction.*: Unsupervised risk estimation [20], [21], [24] and selective prediction [34] show performance is identifiable only under structure. K-Bound differs by certifying the *sign*

of adaptation benefit before committing, with an explicit abstain region and false-adapt control at level α .

III. PROBLEM SETUP AND THEORY

Source $P_S(X, Y)$ has labels at train/validation; target $P_T(X, Y)$ exposes only $P_T(X)$ at test time. Fix loss ℓ , frozen f_0 , and candidate f_a . Target risk $R_T(f) = \mathbb{E}_{P_T}[\ell(f(X), Y)]$. The *adaptation benefit* is

$$\Delta = R_T(f_0) - R_T(f_a), \quad \Delta > 0 \iff \text{adapting helps.} \quad (1)$$

Observable evidence $Z = \phi(X_{1:m}, f_0, f_a)$ is any statistic computable without target labels (entropy, confidence, drift, disagreement, update norms). The central question is when sign Δ is recoverable from Z alone.

What “label-free” means. No target/test labels at deploy; labeled source and held-out validation may calibrate $\hat{\Delta}$ and ε . Evidence Z uses only unlabeled batches, predictions, entropy, drift, and update statistics. The method is **target-label-free**, not label-free of all supervision: labels may be used only in the source/validation calibration split, never in the deployment batch or held-out test decision.

Definition 1 (Regimes). *Knowably helpful*: evidence certifies $\Delta > 0$ (adapt). *Knowably harmful*: evidence certifies $\Delta < 0$ (freeze). *Unknowable*: the same Z is compatible with both signs (abstain).

Lemma 1 (Non-identifiability forces abstention). *There exist worlds P_T^1, P_T^2 with $\text{Law}(Z \mid P_T^1) = \text{Law}(Z \mid P_T^2)$ but $\text{sign } \Delta_1 = -\text{sign } \Delta_2 \neq 0$. No label-free rule can be correct in both; any valid rule must abstain on matched evidence with probability at least $1 - 2\alpha$ when false-adapt and false-freeze are each controlled at α .*

Theorem 1 (K-Bound certificate and benefit-sign frontier). *Suppose $\hat{\Delta}(z)$ satisfies $\mathbb{P}[|\hat{\Delta} - \Delta| \leq \varepsilon] \geq 1 - \alpha$. The rule*

$$g(Z) = \begin{cases} \text{ADAPT} & \hat{\Delta} - \varepsilon > 0, \\ \text{FREEZE} & \hat{\Delta} + \varepsilon < 0, \\ \text{ABSTAIN} & \text{otherwise,} \end{cases}$$

controls false-adapt and false-freeze at level α . On the disagreement region $D = \{x : f_0(x) \neq f_a(x)\}$, writing $M = \mathbb{E}[s \mid D] - \frac{1}{2}$ for a source-calibrated score s of f_a and $\gamma = \mathbb{E}[\eta_a - s \mid D]$ for calibration drift, $\text{sign } \Delta = \text{sign}(M + \gamma)$: the benefit sign is identifiable from label-free evidence iff $|M|$ exceeds the drift budget $|\gamma|$ (exact frontier). Intuitively, label-free heuristics work when drift is negligible and fail when $|\gamma|$ dominates $|M|$ —the obstruction no checkable assumption removes in full generality (Lemma 1; full dichotomy: Appendix A).

Corollary 1 (Invalid evidence $\Rightarrow$ abstain). *When Z is not risk-aligned or $|\hat{\Delta} - \Delta| > \varepsilon$ on the deployment batch, the only valid outcomes are abstention or conservative freeze; committing adapt without a positive certified lower bound violates the target false-adapt level α .*

Algorithm 1 KGA: Knowability-Guided Adaptation

Require: Frozen f_0 ; adapter $\mathcal{A}$; batch $X_{1:m}$; $\hat{\Delta}$, ε , α
Ensure: Decision $g \in \{\text{ADAPT}, \text{FREEZE}, \text{ABSTAIN}\}$

```

1:  $f_a \leftarrow \mathcal{A}(f_0, X_{1:m})$ 
2:  $Z \leftarrow \phi(X_{1:m}, f_0, f_a)$ 
3:  $\hat{\Delta} \leftarrow \hat{\Delta}(Z)$ ;  $L \leftarrow \hat{\Delta} - \varepsilon$ ;  $U \leftarrow \hat{\Delta} + \varepsilon$ 
4: if  $L > 0$  then
5:    $g \leftarrow \text{ADAPT}$ 
6: else if  $U < 0$  then
7:    $g \leftarrow \text{FREEZE}$ 
8: else
9:    $g \leftarrow \text{ABSTAIN}$ 
10: end if
11: return  $g$ 

```

Full proofs, rate theorems, one-bit dichotomy, and validators are in Appendix A.

IV. METHOD: KGA

KGA is the only headline method: a decision wrapper around a fixed adapter, not a new adaptation objective. On a calibration split with labeled pairs (Z, Δ) : fit $\hat{\Delta}(Z)$ (gradient-boosted regressor in all experiments); set ε to the $(1 - \alpha)$ quantile of $|\hat{\Delta} - \Delta|$ on calibration residuals; at deploy extract Z from an unlabeled batch and return adapt, freeze, or abstain (Algorithm 1). Labels enter only through this calibration split; the deployment batch and held-out test scorer never use target labels for adapter, ε , evidence, or rule selection. Synthetic grids use leave-one-task-out calibration; Camelyon17 Protocol G fits on dev hospital seeds and scores held-out test seeds once ($\alpha=0.10$).

On the good event $|\hat{\Delta} - \Delta| \leq \varepsilon$, committing adapt only when $L > 0$ implies $\Delta > 0$ (Theorem 1). Assumptions: exchangeable calibration pairs, bounded benefits, and risk-aligned Z ; no target labels at deploy. Overhead: one forward pass of f_0 and f_a plus lightweight $\hat{\Delta}$ evaluation.

a) *The threshold is derived, not tuned.*: KGA’s commit boundary is not a swept hyperparameter. It is the conformal calibration-drift budget ε that Theorem 1 identifies as the *exact* benefit-sign frontier ($\text{sign } \Delta = \text{sign}(M + \gamma)$), identifiable iff $|M| > |\gamma|$, fixed to the $(1 - \alpha)$ residual quantile on the calibration split and *never* selected on target-test data. Confidence-thresholding (ATC) and agreement-trend selection (AGL) fit a scalar against a labeled or held-out signal and predict *accuracy*; KGA certifies the benefit *sign* with a boundary *forced* by the identifiability budget, and—unlike any thresholded estimator—ABSTAINS on the region where Lemma 1 *proves* no label-free rule can decide. The impossibility construction, not a tuned cutoff, is what separates KGA from “estimate accuracy, then threshold.”

V. EXPERIMENTS

We validate KGA on two controlled corruption/stress tracks (CIFAR-10-C stress, ImageNet-C SAR) and three held-out natural-shift replications (Protocols G, H v2,

candidate	policy	mean acc	regret↓	false-adapt	coverage	beats both?
Tent	always-adapt	0.786	0.0086	—	1.00	
	always-freeze	0.671	0.1232	—	0.00	
	K-Bound	0.793	0.0016	0.00	0.67	yes
EATA	always-adapt	0.798	0.0037	—	1.00	
	always-freeze	0.671	0.1311	—	0.00	
	K-Bound	0.801	0.0015	0.00	0.63	yes
SAR	always-adapt	0.806	0.0018	—	1.00	
	always-freeze	0.671	0.1372	—	0.00	
	K-Bound	0.806	0.0018	0.00	—	ties adapt

TABLE II

CIFAR-10-C STRESS GRID (432 CONDITIONS/METHOD, FIVE SEEDS, $\alpha=0.1$, PROTOCOL A).

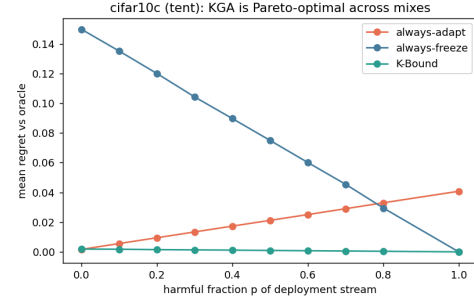


Fig. 1. CIFAR-10-C stress grid: regret-to-oracle vs harmful fraction p per candidate. KGA ties always-adapt at $p=0$ and beats both for $p \gtrsim p^*$.

and M v2; Table I). Calibration and held-out scoring follow Algorithm 2 (Appendix A). Metrics: mean accuracy, regret-to-oracle against the per-condition oracle $g^* = \mathbf{1}[\Delta > 0]$, false-adapt rate (adapt when $B \leq 0$ among committed decisions), and coverage (fraction of non-abstain decisions).

A. CIFAR-10-C stress grid (5 seeds)

The per-corruption suite is helpful-dominated; the *stress* grid crosses severity, batch size, composition, and update aggressiveness (432 conditions/method, ResNet-18, pre-registered Protocol A, seeds 0–4). Harmful base rates are 34% (Tent), 27% (EATA), 16% (SAR). KGA **beats both** trivial policies for Tent and EATA at 0% false-adapt; it ties collapse-resistant SAR (Table II). Among harmful cells KGA adapts 0 and freezes them; among helpful cells it adapts almost all; on marginal cells it abstains. Paired per-condition bootstrap (10^4 resamples, Holm correction over the comparison family) confirms Tent and EATA beat *both* trivial policies after correction; false-adapt is 0/2160 for every candidate. The mixing-ratio Pareto (Fig. 1) shows KGA on the regret-vs-harmful-fraction front, strictly beating both once the harmful fraction exceeds $p^* \approx 0.1$ (Tent/SAR) or 0 (EATA).

B. ImageNet-C SAR (mechanism-faithful)

On ImageNet-C noise (ResNet-50, 36 cells), SAR is harmful on 44% of cells under a mechanism-faithful re-implementation (SAM, recovery reset, EMA) at the learning rate shared across candidates. Always-adapt collapses to 0.319 accuracy while KGA reaches 0.409, **beating both**

candidate	policy	mean acc	regret↓	harmful cells	beats both?
Tent	always-adapt	0.454	0.0007	6%	no (ties adapt)
	always-freeze	0.405	0.0494		
	K-Bound	0.446	0.0082		
EATA	always-adapt	0.444	0.0000	3%	no (ties adapt)
	always-freeze	0.405	0.0386		
	K-Bound	0.440	0.0038		
SAR	always-adapt	0.319	0.1124	44%	yes
	always-freeze	0.405	0.0267		
	K-Bound	0.409	0.0229		

TABLE III

IMAGENET-C, MECHANISM-FAITHFUL SAR (36 CELLS, RESNET-50, $\alpha=0.1$).

TABLE IV

CAMELYON17 PROTOCOL G: CANONICAL KGA, EATA-ONLINE, HELD-OUT TEST ($n=54$).

policy	regret↓	false-adapt	beats both?
always-adapt	0.0013	—	yes
always-freeze	0.0749	—	
K-Bound	3.6×10^{-5}	2.6%	

trivial policies (regret 0.0229 vs 0.1124/ 0.0267). Tent and EATA remain benign; KGA ties always-adapt (Table III). The ImageNet-C result uses our mechanism-faithful SAR reimplementation at a shared learning rate; SAR’s official gentler schedule is left as a robustness check, so the claim is about this harmful-SAR operating point rather than all SAR configurations (Appendix).

C. Natural-shift headline results (Protocols G, H v2, M v2)

On real shifts we separate two layers that must not be conflated: **Headline** = canonical KGA only (Algorithm 1; dev-locked adapter from a pre-registered panel, or fixed adapter for G; GBR $\hat{\Delta}$ + global in-domain ε ; held-out test once). **Diagnostic** = earlier Camelyon/iWildCam runs (sparse Z , multicandidate τ^* , Protocol B/F route audits) that explain failed operating points—Appendix A, §F—and motivated G/H/M v2. Diagnostics are not competing wins. No held-out test labels are used to select the adapter, tune ε , choose evidence features, or revise the decision rule.

D. Camelyon17 Protocol G

WILDS Camelyon17 is a hospital-to-hospital histopathology shift on the composition-stress grid. Composition-stress records use a rich label-free panel; Protocol G is the headline held-out scorer. The **deployed adapter** is EATA-online (fixed). Evidence is the rich label-free panel (17 statistics; harm-AUC 0.95 on the full grid). **Method**: canonical KGA only—GBR $\hat{\Delta}(Z)$ and global ε from dev seeds $\{0, 1\}$, test seeds $\{2, 3, 4\}$ scored once (54 held-out cells). KGA **beats both** trivial policies: regret-to-oracle 3.6×10^{-5} vs always-adapt 0.0013 vs always-freeze 0.075, false-adapt $2.6\% \leq \alpha$, 91% commit (Table IV). Harmful base rate on this adapter is 30%—a mixed real shift, not a freeze-oracle regime.

TABLE V

iWILDCAM PROTOCOL H v2: DEV-LOCKED **TENT_EPISODIC**, HELD-OUT TEST ($n=72$).

policy	regret↓	false-adapt	beats both?
always-adapt	0.1028	—	yes
always-freeze	0.0041	—	
K-Bound	0.0037	0%	

TABLE VI

OFFICE-HOME PROTOCOL M v2: DEV-LOCKED **SAR_ONLINE_AGGRESSIVE**, TARGET-TEST ($n=35$).

policy	regret↓	false-adapt	beats both?
always-adapt	0.0468	—	yes
always-freeze	0.0158	—	
K-Bound	0.0022	0%	

E. iWildCam Protocol H v2

WILDS iWildCam is a geographic camera-trap shift (182 classes). Protocol H v2 dev-locks **tent_episodic** from a six-arm Tent/EATA/SAR panel on id_val (cal seed $\{0\}$, eval seed $\{1\}$), then scores held-out test once (72 cells). Canonical KGA **beats both**: regret 0.0037 vs always-adapt 0.103 vs always-freeze 0.0041, false-adapt $0\% \leq \alpha$ (Table V). Margin vs always-freeze is modest (damage-prevention regime).

F. Office-Home Protocol M v2

Office-Home is a domain shift across art/clipart/product/real. Protocol M v2 dev-locks **sar_online_aggressive** from a six-arm gradient-TTA panel on target-val, then scores held-out target-test (cal seeds $\{0, 1\}$, test seeds $\{0, 1\}$; 35 cells). KGA **beats both**: regret 0.0022 vs always-adapt 0.047 vs always-freeze 0.016, false-adapt $0\% \leq \alpha$ (Table VI). Distinct from the deployed **eata_online_mild** audit (helpful-dominated null; Appendix).

a) *Statistical robustness (bootstrap CIs)*.: Resampling the held-out test conditions ($B=3000$) with the dev calibration held *fixed*—so decisions are never re-tuned on test, and every point estimate reproduces the locked protocol to four decimals—the beats-both is statistically robust on two of the three natural shifts. Office-Home: KGA vs always-freeze +0.014 (95% CI $[0.009, 0.018]$) and vs always-adapt +0.045 ($[0.020, 0.073]$). Camelyon17: vs freeze +0.075 ($[0.057, 0.095]$) and vs adapt +0.0013 ($[0.0006, 0.0020]$). Both gaps’ CIs exclude zero. On iWildCam the margin over always-freeze (+0.0004, $[0.000, 0.0013]$) is within noise, so we report it as *dominate-adapt / tie-freeze*: KGA matches or remains close to the better trivial policy and strictly beats the worse one (vs adapt +0.099, $[0.080, 0.119]$). Per-condition intervals: **research_lock/KBOUND_WIN_BOOTSTRAP_CIS.json**.

b) Mixed-deployment capstone.: Each single dataset is one-sided, so KGA’s simultaneous margin over *both* policies is necessarily small there. Pooling the three held-out shifts’ test conditions into a single heterogeneous deployment ($n=161$; each dataset keeps its own locked dev-calibrated gate, reproducing its single-dataset regret to four decimals) exposes the two-sided regime the certificate is built for: KGA regret 0.0021 versus always-adapt 0.057 ($\sim 27\times$) and always-freeze 0.030 ($\sim 14\times$), with both 95% bootstrap CIs excluding zero (vs freeze $[0.020, 0.037]$, vs adapt $[0.042, 0.067]$) at false-adapt $0.016 \leq \alpha$. When adaptation helps on some conditions and hurts on others, no fixed policy matches the certificate’s per-condition routing (`research_lock/KBOUND_MIXED_STREAM_v1.json`).

c) Supporting and negative regimes.: These tracks support Table I but are not headline policy wins (details: Appendix A). **CIFAR-10-C per-corruption** (65 cells) is helpful-dominated; KGA ties always-adapt. **RxRx1** is harmful-dominated at 1,139-class scale; always-adapt is destructive and KGA matches freeze-oracle safety. **ImageNet-R** remains weak and split-specific; KGA is not yet a stable headline win there. The deployed Office-Home adapter audit remains a helpful-dominated null (Appendix E), distinct from Protocol M v2. **CIFAR-10.1** is low-margin: within-seed calibration ties freeze with 0% false-adapts on Tent/EATA, but cross-seed certificate transfer does not clear the headline bar.

VI. LIMITATIONS AND HONEST SCOPE

Scope of empirical wins. On the stress grid KGA beats both for Tent/EATA and ties SAR (Table II); on ImageNet-C the pattern flips by candidate—Tent/EATA are robust while SAR collapses and KGA beats both under a mechanism-faithful re-run (Table III). Per-corruption CIFAR-10-C and helpful natural shifts are insurance regimes, not policy wins.

Real-shift scope. Headline natural-shift wins are Protocols G, H v2, and M v2 (canonical dev-lock KGA, Tables IV, V, VI). Earlier Camelyon Protocol B/F runs and multicandidate routes are **diagnostics** that motivated these operating points (Appendix); they are not alternate headline wins.

Theory and calibration. The conformal radius assumes exchangeable calibration pairs; sign bracketing without structure is impossible (Theorem 1 and Appendix A). KGA’s value scales with how often catastrophic, detectable harm occurs—not with universal accuracy gains.

Detectability and transfer. ImageNet-R still sits in a weak, split-specific evidence regime (Appendix E). iWild-Cam H v2 clears the certificate bar despite modest margin over always-freeze; intermediate-domain multicandidate routes do not (pooled false-adapt $\gg \alpha$).

Missing baselines and independent validation. ImageNet-C committal gates (ATC-style confidence; Appendix K, Table XVIII) false-adapt at 100% on harmful

SAR cells, while KGA abstains there. Direct POEM-style sequential monitors and additional simple gates on the CIFAR-10-C stress grid remain future checks. Independent replication of ImageNet-R and Office-Home M v2 across more seeds would strengthen stability claims. Abstention reduces coverage by design.

What we do not claim. Universal accuracy gains; adapter selection by test regret; multicandidate routes with violated false-adapt bounds; or wins on helpful-dominated shifts where always-adapt is already oracle-optimal.

VII. REPRODUCIBILITY

All code, locked protocol yamls (`research_lock/`), and result JSONs backing every reported number are in the anonymized submission repository. Controlled and natural-shift headline runs are reproduced via `bash docs/research/kbound/scripts/kbtrain.sh` (protocol-h-v2, protocol-m-v2, stress-grid targets); theory validators and CPU experiments run from the bundled reproduction driver. The core certificate module is packaged as `pip install kbound` (`docs/research/kbound/kbound_pkg/`).

VIII. CONCLUSION

K-Bound reframes test-time adaptation as an identifiability-limited decision problem: adapt when helpful, freeze when harmful, and abstain when label-free evidence cannot support a valid decision. The theory shows that this abstention region is not merely conservative; under matched label-free evidence with opposite adaptation benefit, no valid label-free rule can safely commit. The empirical results support the same regime map: KGA ties or trails only where always-adapt is already near-oracle, but improves over trivial policies when harmful adaptation is frequent, detectable, and costly. These results suggest that future TTA systems should expose not only an adaptation objective, but also a validity layer that decides whether adaptation is knowable before deployment.

APPENDIX

Algorithm 2 is the shared evaluation protocol behind Tables II, IV, V, VI, and the supplementary tracks below.

This appendix is not required for the main empirical claim; it preserves the full theorem stack and validator mapping behind the consolidated Theorem 1. The main paper states a consolidated headline theorem (Theorem 1). Below we give the full five-theorem stack, lemmas, and proof sketches with validator references.

The body is organized as four pillars, with five core theorems serving as the load-bearing statements inside that scaffold. **Pillar I (impossibility floor)**: Theorem 2 plus the Le Cam lower bound and regret identity establish when abstention is information-theoretically forced. **Pillar II (certificate + audit)**: Theorem 3 gives finite-sample false-adapt control, and the one-bit audit machinery of

Algorithm 2 Calibration and held-out evaluation for K-Bound

Require: Conditions $\mathcal{C}$; frozen f_0 ; adapters $\mathcal{A}_1, \dots, \mathcal{A}_K$; calibration split $\mathcal{C}_{\text{cal}}$; test split $\mathcal{C}_{\text{test}}$; level α

Ensure: Regret-to-oracle, false-adapt rate, coverage, beats-both verdict

```

1: for all  $c \in \mathcal{C}_{\text{cal}}$  do
2:   for all adapters  $\mathcal{A}_k$  do
3:      $f_a^{(k,c)} \leftarrow \mathcal{A}_k(f_0, X_c)$ 
4:      $Z_{k,c} \leftarrow \phi(X_c, f_0, f_a^{(k,c)})$ 
5:      $\Delta_{k,c} \leftarrow R_c(f_0) - R_c(f_a^{(k,c)})$ 
6:   end for
7: end for
8: Train  $\hat{\Delta}(Z)$  on calibration pairs  $(Z_{k,c}, \Delta_{k,c})$ 
9:  $\varepsilon \leftarrow (1 - \alpha)$  empirical quantile of  $\{|\hat{\Delta}(Z_{k,c}) - \Delta_{k,c}|\}$ 
10: for all  $c \in \mathcal{C}_{\text{test}}$  do
11:   for all adapters  $\mathcal{A}_k$  do
12:     Run Algorithm 1 with  $(f_0, \mathcal{A}_k, X_c, \hat{\Delta}, \varepsilon)$ 
13:     Record decision, regret-to-oracle, coverage, false-adapt indicator
14:   end for
15: end for
16: Aggregate metrics vs. always-adapt, always-freeze, per-condition oracle

```

Theorem 5(iv) supplies a common falsifier for the legacy budget premises (formerly presented as separate propositions, now treated as corollaries of one audit framework).

Pillar III (identifiable regimes): Theorem 4 and Theorem 5 characterize exactly what label-free evidence identifies and what one-bit supplement remains irreducible.

Pillar IV (rates): Theorem 6 gives matching evidence-channel rates. The open piece—a weakest falsifiable class on which one declared bit suffices—is settled conditionally in Appendix N. Supporting variants (multiclass/regression reductions, label-shift and drift boundaries, ambiguity-reach unification, router capacity, anytime certificates, the γ -meter, and the agreement–accuracy law) are presented as corollaries or companions in the companion manuscript, each with a validator; every numeric claim here traces to a script in the public repository.

Lemma 2 (Reduction to the disagreement region). *Let $D = \{x : f_0(x) \neq f_a(x)\}$ (observable). For binary Y and 0/1 loss, with $\bar{a} = \mathbb{P}_T(f_a=Y \mid D)$, $\Delta = 2\mu(D)(\bar{a} - \frac{1}{2})$, so for $\mu(D) > 0$: $\text{sign } \Delta = \text{sign}(\bar{a} - \frac{1}{2})$. Writing s for any source-calibrated score of f_a , $M := \mathbb{E}_{\mu|D}[s] - \frac{1}{2}$ (observable) and $\gamma := \mathbb{E}_{\mu|D}[\eta_a - s]$ (unobservable), $\boxed{\text{sign } \Delta = \text{sign}(M + \gamma)}$. For K -class 0/1 loss $\Delta = \mu(D)(p_a - p_0)$, and for squared loss the sign reduces to one moment on D ; both extend the identity verbatim (manuscript, App. C).*

Proof sketch. Off D the losses coincide; on D exactly one of f_0, f_a is correct (binary), giving $\mathbb{E}[\delta \mid D] = 2\bar{a} - 1$; linearity splits $\bar{a} - \frac{1}{2} = M + \gamma$. Validated on synthetic draws (identity

exact to machine precision; multiclass to 10^{-16} over 4000 trials). $\square$

Lemma 3 (Plug-in regret identity). *For any estimator $\hat{\Delta}$, the gate $\hat{g} = \mathbf{1}[\hat{\Delta} > 0]$ satisfies $R(\hat{g}) - R(g^*) = \mathbb{E}[\Delta(Z) \mathbf{1}\{\text{sign } \hat{\Delta} \neq \text{sign } \Delta\}]$ with $g^* = \mathbf{1}[\Delta > 0]$ Bayes; on $\{|\Delta| < \varepsilon\}$ committal regret is $\leq \varepsilon$, justifying abstention in the low-margin band. Validated to machine precision on synthetic regret identities.*

Proposition 1 (Finite-sample minimax regret floor for label-free gating). *Intuition. The identity of Lemma 3 is an equality, not a hardness statement; pairing it with Le Cam’s two-point method turns it into a finite- n lower bound on the regret of any label-free gate, which does not vanish as $n \rightarrow \infty$ when the two worlds are rescaled to stay equally hard. Let $\{P_\theta\}_{\theta \in \{-1, +1\}}$ be two target worlds sharing the same disagreement structure with constant benefit magnitude $|\Delta(P_\theta)| \equiv \Lambda > 0$ and opposite signs $\text{sign } \Delta(P_\theta) = \theta$, and let $Q_\theta^{(n)}$ denote the law of any label-free evidence Z computed from n unlabeled observations under P_θ . Then for every label-free committal gate $\hat{g} : Z \mapsto \{\text{ADAPT}, \text{FREEZE}\}$ and every finite n ,*

$$\inf_{\hat{g}} \max_{\theta \in \{\pm 1\}} [R_{P_\theta}(\hat{g}) - R_{P_\theta}(g^*)] \geq \frac{\Lambda}{2} \left(1 - \text{TV}(Q_{-1}^{(n)}, Q_{+1}^{(n)})\right) \geq \frac{\Lambda}{4} e^{-\text{KL}(Q_{-1}^{(n)} \| Q_{+1}^{(n)})}$$

In particular there is an explicit Gaussian two-point family $(X \sim \mathcal{N}(\theta d_n, 1))$ on D , $f_0 = \mathbf{1}[x > 0]$, $f_a = \mathbf{1}[x < 0]$, with the target label rule giving $\Delta = \theta$, so $\Lambda = 1$) for which, taking $d_n = c/\sqrt{n}$, both bounds are constant in n : $\text{TV}(Q_{-1}^{(n)}, Q_{+1}^{(n)}) = 2\Phi(c) - 1$ and the floor equals $\Lambda \Phi(-c)$, with the Bretagnolle–Huber certificate $\frac{\Lambda}{4} e^{-2c^2} > 0$. Theorem 2 is the $c \rightarrow 0$ ($\text{TV} \rightarrow 0$) face, where the floor saturates at $\Lambda/2$.

Proof. By Lemma 3, since $|\Delta| \equiv \Lambda$ and g^* is the constant correct action $\text{sign } \theta$ in world θ , $R_{P_\theta}(\hat{g}) - R_{P_\theta}(g^*) = \Lambda Q_\theta^{(n)}(\hat{g} \text{ commits to the wrong sign})$. Averaging over the uniform prior on θ gives $\frac{1}{2} \sum_\theta [R_{P_\theta}(\hat{g}) - R_{P_\theta}(g^*)] = \frac{\Lambda}{2} M(\hat{g})$, where $M(\hat{g}) = Q_{+1}^{(n)}(\hat{g}=\text{FREEZE}) + Q_{-1}^{(n)}(\hat{g}=\text{ADAPT})$ is the mixed committal error of Theorem 2(ii). Le Cam’s testing-affinity identity yields $\inf_{\hat{g}} M(\hat{g}) = 1 - \text{TV}(Q_{-1}^{(n)}, Q_{+1}^{(n)})$, and $\max \geq \text{avg}$ gives the minimax form; Bretagnolle–Huber ($1 - \text{TV} \geq \frac{1}{2} e^{-\text{KL}}$) gives the closed-form certificate. For the Gaussian family the sufficient statistic $\bar{X} \sim \mathcal{N}(\theta d_n, 1/n)$ has $\text{TV} = 2\Phi(\sqrt{n} d_n) - 1$ and per-observation $\text{KL} = 2d_n^2$, so $d_n = c/\sqrt{n}$ makes both n -independent; $|\Delta| = 1$ exactly because each label rule makes one of f_0, f_a correct on a probability-one event. Validated (`val_thm2_lecam_finite_n.py`, seeds fixed): across $c \in \{0.25, 0.5, 1, 1.5\}$ and $n \in \{10, 50, 200, 1000\}$ the empirical floor is constant in n , the Bayes (sample-mean threshold) gate meets it (so it is tight), and a brute-force minimax over a panel of label-free rules (all thresholds/polarities on five statistics) never beats it within Monte-Carlo error. $\square$

Theorem 2 (The unknowable regime is real and tight). *Intuition. Identical label-free evidence can support opposite adaptation benefits; any valid rule must then abstain rather than guess. (i) There exist target worlds P_T^1, P_T^2 with identical evidence laws, $\text{Law}(Z | P_T^1) = \text{Law}(Z | P_T^2)$, and $\text{sign } \Delta_1 = -\text{sign } \Delta_2 \neq 0$ (explicit witness: $X \sim \mathcal{N}(0, 1)$, $f_a = 1 - f_0$, flipped labels). No label-free rule distinguishes them; the worst-case-optimal committal-regret action is ABSTAIN. (ii) Quantitatively, for any committal rule g the mixed error obeys the Le Cam identity $\inf_g M(g) = 1 - \text{TV}(P_{T,Z}^{1,(n)}, P_{T,Z}^{2,(n)})$, equal to 1 in the witness for every n and decaying only as the evidence laws separate. (iii) Any rule with false-adapt and false-freeze rates $\leq \alpha$ must abstain on matched evidence with probability $\geq 1 - 2\alpha$: abstention is mandatory, not conservative.*

Proof sketch. (i) The witness makes Z a function of X alone with $P^1(X) = P^2(X)$. (ii) A committal rule is a test between the two evidence laws; Le Cam’s testing affinity plus data processing. (iii) Matched evidence forces common action probabilities q_a, q_f ; validity caps each at α . Validated: empirical $\inf_g M$ tracks $1 - \text{TV}$ across all n , 100% abstention on the witness. $\square$

Theorem 3 (The adapt/freeze/abstain certificate). *Intuition. When a benefit estimate carries a valid uncertainty radius, committing only outside the abstain band controls false-adapt and false-freeze at level α . Suppose $\hat{\Delta}(z)$ admits a radius $\varepsilon(z)$ with $\mathbb{P}[|\hat{\Delta} - \Delta| \leq \varepsilon] \geq 1 - \alpha$. The rule ADAPT if $\hat{\Delta} - \varepsilon > 0$, FREEZE if $\hat{\Delta} + \varepsilon < 0$, else ABSTAIN, has false-adapt and false-freeze rates $\leq \alpha$ each. With $\hat{\Delta}$ an empirical-Bernstein mean of n bounded paired benefits (variance proxy σ^2 , range $b - a$), the certificate commits once $n \gtrsim (2\sigma^2/\Delta^2) \log(2/\alpha) + 3(b - a)|\Delta|^{-1} \log(2/\alpha)$: sample complexity scales as Δ^{-2} , logarithmically in $1/\alpha$. A sequential, anytime-valid e-process version (valid at any stopping time; β -drift-robust variant included) is Theorem B.1 of the manuscript.*

Proof sketch. On the good event, ADAPT implies $\Delta \geq \hat{\Delta} - \varepsilon > 0$; the complement has mass $\leq \alpha$; commitment when $\varepsilon(n) < |\Delta|$ inverts the Bernstein radius. The residual burden — a calibration sample exchangeable with deployment — is exactly what Theorem 5(iv) makes auditable. Validated on synthetic e-process tests (false-adapt $\leq \alpha$ at nominal level). $\square$

Theorem 4 (Exact benefit-sign frontier). *Intuition. On the disagreement region, the benefit sign is knowable iff an observable margin $|M|$ exceeds a calibration-drift budget β ; otherwise abstention is forced. Fix the observables (μ, P_S, f_0, f_a) with $\mu(D) > 0$, so M is determined, and let $\mathcal{C}_\beta = \{P_T : |\gamma| \leq \beta\}$. The statements below hold unconditionally: for every drift budget $\beta \geq 0$ and every target with $\mu(D) > 0$, with β a free parameter of the frontier rather than an assumption (and (iii) shows every identifying class is of this form). (i) γ is a one-dimensional*

sufficient statistic for $\text{sign } \Delta$ given the observables. (ii) $\text{sign } \Delta$ is identifiable over $\mathcal{C}_\beta$ iff $|M| > \beta$, and then equals $\text{sign } M$; the minimax committal error is 0 if $|M| > \beta$ and $\frac{1}{2}$ otherwise, so the three-way rule (ADAPT if $M > \beta$, FREEZE if $M < -\beta$, else ABSTAIN) is the unique maximal sound certificate. (iii) Necessity: any class on which $\text{sign } \Delta$ is identifiable lies on one side of the hyperplane $\gamma = -M$; every identifying assumption is a one-sided constraint on γ . (iv) The label-free heuristics ATC [13], DoC, GDE [18], COT [19], AETTA [17], and agreement-on-the-line [15], [16] are exactly the $\beta=0$ face: sound iff $\gamma = 0$, sharing the single failure mode $\gamma \neq 0$.

Proof sketch. All from Lemma 2: (ii) a $\pm\beta$ perturbation crosses 0 iff $|M| \leq \beta$, plus the two-point argument of Theorem 2; (iii) constancy of $\text{sign}(M + \gamma)$ is one-sidedness; (iv) each heuristic thresholds an observable surrogate, i.e. commits to $\text{sign } M$. Validated: frontier law exact over 5×10^5 draws; the unconditional family characterization (identifiable over $\mathcal{C}_\beta$ iff $|M| > \beta$) holds for every tested β ; ATC error matches the predicted event to machine precision at drift 0.04–0.28 (synthetic validators for Theorems T1–T5 and AGL). $\square$

Corollary 2 (Knowable regimes). (a) *Under covariate shift ($P_S(Y|X) = P_T(Y|X)$, bounded density ratio), importance weighting identifies Δ and its sign from labeled source plus unlabeled target (γ -budget discharged with $\beta = 0$ on the importance-weighted score); the premise itself is not label-free testable, re-entering Theorem 2 when it fails. (b) Conversely the budget is irreducible: no label-free certificate identifies $\text{sign } \Delta$ without a supplied bound on γ — the assumption-free certificate sought by prior heuristics does not exist. (Label shift, bounded drift, smooth drift, and the unified ambiguity-reach hierarchy: manuscript, App. C.)*

Theorem 5 (One-bit dichotomy; Conjecture 1 resolved). *Intuition. Label-free observables fix every adaptation statistic except one global orientation bit; no testable assumption class removes that bit in full generality. Consider a panel $f_0, f_1, \dots$ on D with correctness indicators conditionally independent given Y (CEI) and per-class symmetric accuracies (jointly, hypothesis $\mathbf{H}$; advantages $b_j = 2a_j - 1$, agreements $c_{ij} = 2A_{ij} - 1$; the rank-one algebra and anchor are classical [25], [26], [28], [29] — new here are the deficit, the dichotomy, and the audit). (i) H is minimal: under CEI alone, $c_{ij} - b_i b_j = \pi(1 - \pi)\delta_i \delta_j$ exactly (δ_j the per-class accuracy gap), so the rank-one law holds for a $K \geq 3$ panel iff every $\delta_j = 0$. (ii) One bit: under H the full evidence law determines every $|b_j|$, every product $b_i b_j$, and every benefit magnitude $|b_j - b_0|$ — everything except a single global flip $b \mapsto -b$, which is free at every moment order: the label complement preserves the entire evidence law ($\text{TV} = 0$) while negating every benefit sign. The drift $\gamma = b_a/2 - M$ is thereby point-identified up to that bit. (iii) No testable bracketing (the dichotomy): in any output space (binary, multiclass, regression) there is a swap*

involution $P \mapsto P^*$ — conditional mass exchange between the two disagreeing predictions on D , midpoint reflection for regression — with $\text{TV}(L(P), L(P^*)) = 0$ for the law L of all label-free observables (labeled source included) and $\Delta(P^*) = -\Delta(P)$. Hence no evidence-definable (label-free testable) assumption class identifies $\text{sign } \Delta$; every identifying assumption is a bit-selector (γ -budgets select $\text{sign } M$; majority-above-chance selects $\text{sign}(\text{median } b)$; the agreement-line slope selects $\text{sign}(1 - 2\bar{w})$); and the minimal untestable supplement is exactly one bit, which suffices under H . (iv) Budgets are falsifiable, never verifiable: the bit-robust test rejecting “ $|\gamma| \leq \beta$ ” when $\min_{\text{flip}} |\hat{\gamma}| > \beta + r_n$ is level- α with power $\geq 1 - \alpha$ beyond $2r_n$; verification fails exactly on the blind set $\{ ||M| - |b_a|/2| \leq \beta < |M| + |b_a|/2 \}$, and the $M \geq 4$ rank-one residual τ is sound but not complete (strictly interior stealth laws with $\tau = 0$ exist that bias b and can flip the recovered decision).

Proof sketch. (i) Expand c_{ij} over Y and cancel cross terms. (ii) Triple products $c_{ik}c_{il}/c_{kl} = b_i^2$; the complement $Y' = 1 - Y$ fixes predictions (maps of X) hence the whole evidence law, while $b \mapsto -b$ and H is preserved. (iii) The swap preserves the X -marginal and source, so L is fixed at every order; on D it exchanges $p_a \leftrightarrow p_0$ (or negates $f_0 + f_a - 2Y$), so Δ negates; identifiability on a swap-closed class is impossible, and selecting one member per swap orbit is by definition one bit. Algebraically: observables generate the *even* correctness subalgebra ($s_i s_j = u_i u_j$), the decision is *odd* ($s_j = u_j v$), and the swap is $v \mapsto -v$. (iv) Empirical-Bernstein for $\hat{M}$, a product-ratio perturbation bound for $\hat{b}_a$, and the flip-min for bit-robust soundness; the stealth construction is a strictly interior LP on the 2^4 correctness patterns matching all pairwise moments to a rank-one target with a shifted first moment. Validated (machine-checked, seeds fixed): deficit matches theory to 5.7×10^{-4} ; flip $\text{TV} = 0$ with $\max_j |b'_j + b_j| = 1.1 \times 10^{-16}$; swap evidence change exactly 0 with $\Delta^* = -\Delta$ (multiclass $K=5$ and regression); audit level $0.000 \leq \alpha=0.05$, power 1.000; stealth $\tau = 2 \times 10^{-17}$ with recovery bias 0.229 (Figs. 2–5). Full proofs appear in the supplementary theory appendix. $\square$

Conjecture 1 (Label-free bracketing — **resolved**). *The unconditional label-free bracketing of the ordinal comparison on D (posed as the open piece in earlier versions) does not exist: by Theorem 5(iii) no evidence-definable assumption identifies $\text{sign } \Delta$, and the minimal untestable supplement is one bit. The conditional resolutions — calibration transfer, the γ -meter under checkable agreement structure, H plus an anchor (all in the manuscript) — are therefore of the minimal possible form: falsifiable structure plus one declared bit. On the margin-monotone relative-calibration class $\mathcal{C}_{\text{mono}}$ (Definition 2) one declared bit certifies $\text{sign } \Delta$ unconditionally (Theorem 7(i)–(ii)); and $\mathcal{C}_{\text{mono}}$ is a weakest such class under a general-position assumption (Theorem 7(iii)). The unconditional weakest class (general position removed) remains open (Remark 1).*

Main-paper result	Demoted to manuscript (with validators)
Thm 2	forced-abstention, minimax-floor variants
Thm 3	anytime e-process; β -drift; router capacity
Thm 4	label-shift / drift boundaries; reach hierarchy
Thm 5	γ -meter; invisible reach; multiclass H
Thm 6	labeled-channel rates; AGL law and failures

TABLE VII

THE FIVE-THEOREM CONSOLIDATION. FIFTY-PLUS PRIOR NUMBERED RESULTS SURVIVE AS COROLLARIES OR MANUSCRIPT SECTIONS; NONE WAS DELETED, AND EVERY ONE KEEPS ITS MACHINE-CHECKED VALIDATOR.

Theorem 6 (The evidence-channel rate: labels buy constants and the bit, not the rate). *Intuition. Unlabeled agreement statistics and labeled paired benefits share the same $1/\sqrt{m}$ rate; labels change constants and supply the missing bit, not the exponent. Within audited H with margins $(\min_{i \neq j} |c_{ij}| \geq c_{\min}, \min_j |b_j| \geq b_{\min})$ and a declared bit, m unlabeled samples on D give $|\hat{b}_j - b_j| \leq C \sqrt{\log(K/\delta)/m}$ with $C = \Theta(1/(b_{\min} c_{\min}^2))$, so $\text{sign}(b_j - b_0)$ is certified at $m \asymp C^2 (b_j - b_0)^{-2} \log(K/\delta)$; a Le Cam two-point bound inside H shows $m^{-1/2}$ is minimax for the evidence channel. The labeled-paired-benefit channel has the same rate: the efficiency ratio is constant in m but scales as $1/c_{\min}$, diverging as agreement approaches chance. Relatedly, accuracy is an exact affine function of agreement-with-reference iff the disagreement win rate is constant (slope $1 - 2\bar{w}$), the anchored form of agreement-on-the-line; its slope sign is the same bit as in Theorem 5(iii) (manuscript, §AGL).*

Proof sketch. Hoeffding on agreements, the product-ratio perturbation lemma, and a χ^2/KL bound between K -bit pattern laws differing by ε in one coordinate ($\text{KL} \leq C'\varepsilon^2$), tensorized; Bretagnolle–Huber finishes. Validated: recovery slope -0.51 ; minimax slope -0.525 tracking $c_0/\sqrt{m}$; $\text{KL}/\varepsilon^2 \in [0.164, 0.171]$; efficiency $2.15 \rightarrow 4.49 \rightarrow \infty$ as $c_{\min} 0.21 \rightarrow 0.067 \rightarrow \leq 0.022$ (Fig. 4; AGL: $R^2 = 1.000$ under constant w , slope $1 - 2w$ exact on synthetic checks). $\square$

The weakest one-bit class. Theorem 5 fixes the supplement at one bit; the *weakest falsifiable class* on which that bit suffices is settled conditionally — a margin-monotone relative-calibration class on which one declared bit certifies $\text{sign } \Delta$ unconditionally, and a weakest such class under a general-position assumption (Appendix N, Thm 7). The *unconditional* weakest class remains open.

A. Validation figures

This appendix holds **non-headline** tracks: helpful-dominated controls, honest nulls, and the Camelyon/iWildCam **diagnostic ladder** (Protocols B/F, multicandidate routes, sparse- Z failures) that motivated the headline Protocols G / H v2 / M v2 in the main text.

B. Anomaly routing (62 tasks)

On 62 label-free anomaly-detection tasks (85% harmful), KGA makes 0 false-adapt decisions and matches always-freeze (regret 0.0055 vs always-adapt 0.051).

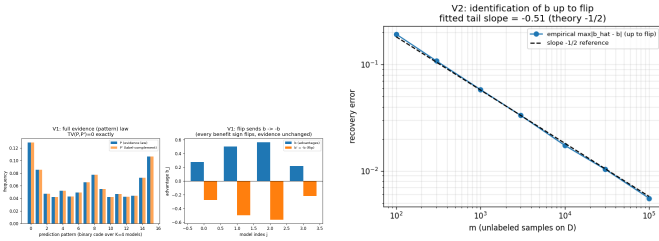


Fig. 2. One-bit identification (Synthetic validator; Thm. 5(ii)).

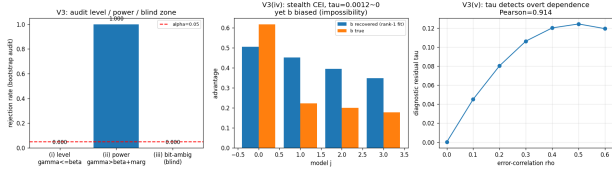


Fig. 3. Budget audit (Synthetic validator; Thm. 5(iv)).

C. Online non-stationary CIFAR-10

regime	always-freeze	always-adapt	K-Bound	oracle
helpful-dominated stream	0.472	0.548	0.553	0.704
harm-dominated stream	0.440	0.339	0.426	0.630
mean across regimes	0.456	0.444	0.490	0.667

TABLE VIII

ONLINE CIFAR-10 TTA (3 SEEDS/REGIME). KGA IS NEAR-BEST IN BOTH REGIMES.

D. CIFAR-10-C per-corruption (helpful-dominated control)

method	mean acc (65 cells)	false-adapt	regret-to-oracle
frozen	0.412	0.000	0.217
Tent (online)	0.626	0.015	0.0030
K-Bound (KGA)	0.626	0.015	0.0032
oracle (per-cell)	0.629	—	—

TABLE IX

PER-CORRUPTION CIFAR-10-C: KGA TIES ALWAYS-ADAPT (SAFETY INSURANCE).

E. Excluded wins and regime boundaries

Several benchmarks flag beats-both under alternate routes or adapters but do *not* qualify as headline natural-shift wins under the Protocol G/H/M v2 bar (canonical KGA, dev-locked adapter from pre-registered panel, in-domain calibration, held-out test, false-adapt $\leq \alpha$).

a) *Superseded protocols.*: iWildCam Protocol H v1 (fixed `sar_online`) and OfficeHome Protocol M v1 (14-candidate win-finder) are audit-only; headline claims use H/M v2.

b) *Office-Home (deployed adapter).*: The source-selected deployed adapter (mild online EATA) is *helpful-dominated* on target test (0% harmful cells; always-adapt is oracle). KGA has 0% false-adapts but does not beat both (regret 0.015 vs always-adapt 0). An aggressive online SAR arm can beat both on the same test split, but that arm

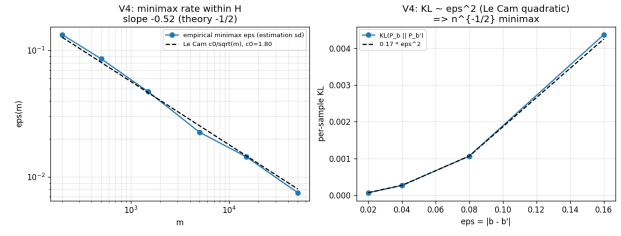


Fig. 4. Evidence-channel rate (Synthetic validator; Thm. 6).



Fig. 5. Conjecture 1 closure validator.

was not the deployed adapter; reporting it as a win would be adapter shopping. The multicandidate deploy route abstains on 100% of target-test cells (certificate transfer-AUC ≈ 0.49). Protocol M v2 (main text) uses a separate dev-lock panel and held-out target-test scorer.

c) *ImageNet-R.*: The diverse-backbone panel is helpful-leaning (mean $\bar{B} = +0.017$, harmful rate 13.9%, best harm-AUC 0.662). The multicandidate route abstains on all 48 conditions; single-candidate KGA often ties always-adapt (oracle) on helpful backbones. This is the weak-detectability / unknowable regime predicted by Corollary 2.

d) *iWildCam intermediate domains.*: On `id_val`, a multicandidate router can flag beats-both on subsets, but pooled false-adapt on committed adapts is $\approx 78\% \gg \alpha$ —disqualified. Headline claim: Protocol H v2 on held-out test only (Table V).

e) *Camelyon17 four-seed hospital slice.*: The original four held-out hospital seeds are helpful-dominated (0% harmful); KGA beats always-freeze but not always-adapt (Table X). Protocol G uses a composition-stress grid with rich evidence where harmful cells are mixed (30%) and detectable—a different regime on the same benchmark family.

f) *CIFAR-10.1.*: Within each seed, Tent/EATA abstain heavily with 0% false-adapts and tie always-freeze. Cross-seed calibration (fit on seeds 0–2, test on 3–4) does *not* clear the headline bar (false-adapt 44%); low-margin benefits do not transfer across independent stream seeds. Report CIFAR-10.1 as a within-seed safety probe, not a fourth natural headline win.

F. Camelyon17 diagnostic ladder

a) *Protocol F route audit (diagnostic only—not headline).*: Protocol G (main-text headline) uses canonical GBR + global ε on EATA-online only. The table below audits *alternate routes* on the same GPU records: multicandidate routing **fails**; pooled GBR+global and PPI+Mondrian

candidate	policy	mean balanced acc	regret↓	beats both?
Tent	always-adapt	0.817	0.000	no
	always-freeze	0.786	0.031	
	K-Bound	0.797	0.020	
EATA	always-adapt	0.836	0.000	no
	always-freeze	0.786	0.050	
	K-Bound	0.818	0.018	

TABLE X

WILDS CAMELYON17 (4 SEEDS): HELPFUL-DOMINATED; KGA BEATS FREEZE, NOT ADAPT.

can beat both but are not the headline method. Sparse-evidence Protocol B also fails (false-adapt $0.33 \gg \alpha$). These negatives motivated G.

TABLE XI

Diagnostic only. PROTOCOL F ROUTE AUDIT (SAME RECORDS AS PROTOCOL G).

variant	regret↓	false-adapt	beats both?
multicandidate router	0.093	71%	no
GBR + global ϵ	0.0019	3.3%	yes
PPI + Mondrian	0.0018	3.3%	yes

G. RxRx1 (harmful, freeze-oracle)

RxRx1 9+ grid (360 conditions)	value
harmful base rate	81.5%
mean benefit $\bar{B}$	-0.063
harm detectability (best AUC)	0.78–0.84
SAR-online always-adapt (collapse)	0.024
KGA false-adapts	0

TABLE XII

RxRx1: KGA MATCHES FREEZE-ORACLE; DAMAGE PREVENTION, NOT BEATS-BOTH.

H. ImageNet-R (unknowable regime)

ImageNet-R (48 condition–seed pairs)	value
harmful base rate	13.9%
mean benefit $\bar{B}$	+0.017
harm detectability (best AUC)	0.662
KGA decisions	48/48 abstain

TABLE XIII

IMAGENET-R: MANDATORY ABSTENTION (COROLLARY 2).

I. Office-Home and iWildCam

J. CIFAR-10.1 (low-margin natural shift, Protocol K audit)

Over five seeds, KGA abstains heavily on Tent/EATA (59–86%), makes zero false adapts there under *within-seed* calibration, and ties always-freeze while always-adapt overpays; SAR is mostly helpful (5% harmful). Cross-seed transfer of the certificate (Protocol K) does not meet the headline bar; see §E.

Office-Home (gradient-TTA)	val	test
Harmful base rate	0.17	0.12
Certificate transfer-AUC	0.33	0.49
KGA false-adapts / beats both	0 / no	0 / no

TABLE XIV

OFFICE-HOME: ZERO FALSE-ADAPTS; NO BEATS-BOTH (INVALID TRANSFER EVIDENCE).

iWildCam	val (multicand)	test
Harmful base rate	59%	83%
Best harm-AUC	≈ 0.62	0.69
Multicandidate beats both?	no	no
Protocol H v2 canonical KGA (tent_episodic)	—	yes (0% FA)

TABLE XV

iWILDCAM: **diagnostic** MULTICANDIDATE ROUTE ON VAL/TEST VS **headline** PROTOCOL H V2 CANONICAL KGA ON TEST (TABLE V IN MAIN TEXT).

K. ImageNet-C diagnostics

An earlier ImageNet-C SAR port (31% harmful) motivated the mechanism-faithful re-run (Table III in the main text). Decision baselines and α /evidence ablations appear in the frontier appendix (App. K).

All experiments are reproducible from the anonymized submission repository: theory validators, pre-registered Protocols A/G/H v2/M v2, and locked result JSONs (experiments/kbound/results/, research_lock/KBOUND_HEADLINE_FINDINGS.json). Environment: Python 3.12 and PyTorch 2.5.1 for GPU runs; `kbtrain.sh` wraps venv activation and natural-shift protocol scorers. Core certificate code: `pip install -e docs/research/kbound/kbound_pkg/`.

Track	Repro command / artifact	Status
CIFAR-10-C stress (5 seeds)	<code>kbtrain.sh stress targets</code>	locked JSON
ImageNet-C SAR faithful	result JSON + Table III	locked
Camelyon17 Protocol G	<code>analyze_F</code> on protocol G bundle	locked
iWildCam Protocol H v2	<code>kbtrain.sh protocol-h-v2</code>	locked
Office-Home Protocol M v2	<code>kbtrain.sh protocol-m-v2</code>	locked
Theory validators	<code>theory_validation/</code> suite	pass

TABLE XVI

REPRODUCIBILITY STATUS FOR HEADLINE TABLES (JUNE 2026 LOCK).

Every theorem has a numerical sanity-check in the theory-validation suite; multi-seed experiments report pooled means with paired bootstrap where pre-registered.

The benefit-sign frontier (Thm. 4) predicts that on the low-margin band any *committal* label-free rule must commit sign errors, whereas a valid certificate must *abstain* there. This appendix demonstrates that prediction on the two *synthetic-corruption* streams whose operating point the certificate is calibrated on (CIFAR-10-C, ImageNet-C); it makes *no* external-validation claim. The honest natural-shift (Camelyon17) finding—a limitation, not a win—closes the appendix (App. N).

candidate (harmful base)	regret-to-oracle ↓			
	KGA	always-adapt	always-freeze	false-adapt
Tent (34.5%)	0.0016	0.0086	0.1232	0.00
EATA (27.1%)	0.0015	0.0037	0.1311	0.00
SAR (15.7%)	0.0018	0.0018	0.1372	0.00

TABLE XVII

CIFAR-10-C stress grid (432 conditions/candidate, $\alpha=0.1$).
KGA BEATS BOTH FOR TENT AND EATA AT A 0% FALSE-ADAPT RATE, AND TIES ALWAYS-ADAPT FOR SAR.

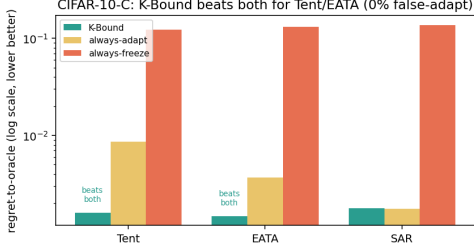


Fig. 6. **CIFAR-10-C frontier (regret-to-oracle, log scale).** KGA is below *both* always-adapt and always-freeze for Tent and EATA at 0% false-adapt, and ties always-adapt for SAR (Table XVII).

L. CIFAR-10-C: K-Bound beats both trivial policies at zero false-adapt

On the pre-registered CIFAR-10-C stress grid (432 conditions/candidate), KGA adapts the helpful conditions and freezes the harmful ones with a *zero* false-adapt rate, so it strictly beats both always-adapt and always-freeze for the two benign candidates (Tent, EATA) and ties the collapse-resistant SAR (Table XVII).

M. ImageNet-C/SAR: the frontier on real corruptions—committal rules fail, K-Bound does not

On the logged ImageNet-C conditions (36 cells/candidate; SAR is harmful on 44.4% of cells), every *committal* label-free heuristic must place its commits somewhere on the low-margin band and therefore false-adapts: a plug-in **ATC** [13] confidence rule adapts into harmful cells on 100% of its commits (false-adapt 1.00, harmful-cells-adapted 0.25), and even its leave-one-out-tuned variant keeps false-adapt = 1.00. Among the rules compared, KGA reaches zero false-adapt and zero harmful-cells-adapted by *abstaining* on the band (coverage 0.39), consistent with the frontier prediction (Table XVIII).

N. Historical note (Camelyon17, frozen synthetic threshold)

Under a **frozen, Camelyon-free** multicandidate threshold $\tau^*=0.52$ (calibrated only on synthetic grids), the certificate does *not* beat both trivial policies on an early Camelyon17 composition grid (regret KGA 0.076 vs always-freeze 0.071). That failure motivated in-domain calibration and rich evidence; the headline result is Protocol G (main text, Table IV), which uses dev-hospital calibration and does not transplant τ^* from synthetic runs.

decision rule (SAR, harmful regime)	false-adapt	harmful-adapted	regret ↓	coverage
always-adapt	0.44	1.00	0.1124	1.00
always-freeze	—	0.00	0.0267	1.00
ATC-style confidence (plug-in)	1.00	0.25	0.0673	1.00
ATC-style confidence (LOO)	1.00	0.06	0.0375	1.00
K-Bound (KGA)	0.00	0.00	0.0229	0.39

TABLE XVIII

Committal label-free heuristics false-adapt exactly where KGA abstains (ImageNet-C, REFERENCE-PARITY SAR, 36 CELLS, $\alpha=0.1$). ATC COMMITTS ONTO THE LOW-MARGIN BAND AND FALSE-ADAPTS ON 100% OF ITS COMMITS; KGA ABSTAINS THERE (22/36 ABSTAIN) FOR 0 FALSE-ADAPTS. ON THE BENIGN CANDIDATES KGA ALSO KEEPS 0 FALSE-ADAPTS AT HIGHER COVERAGE (TENT: REGRET 0.0060, COVERAGE 0.69; EATA: REGRET 0.0047, COVERAGE 0.75).

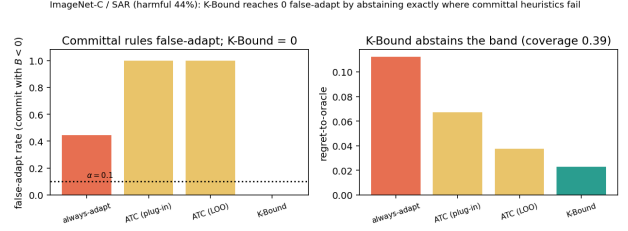


Fig. 7. **ImageNet-C/SAR frontier.** Committal label-free rules false-adapt on the harmful regime; KGA reaches zero false-adapt by abstaining the low-margin band (coverage 0.39; Table XVIII).

The weakest one-bit class (resolving the open piece of Conjecture 1). Theorem 5 fixes the supplement at *one bit*; what remained open is the *weakest falsifiable class* on which that bit suffices. We settle it conditionally. On D write $a(x) := 2\eta_a(x) - 1$ with $\eta_a(x) = \mathbb{P}_T(f_a=Y | x)$, so $\Delta = \int_D a d\mu$ (Lemma 2). Let $m(x) := 2s(x) - 1$ be the observable relative (calibrated-score) margin, so $M = \frac{1}{2}\mathbb{E}_{\mu|D}[m]$, and let the observable *relative-calibration field* be $c(x) := w(x)(m(x) - m^*)$ with observable nonnegative weight $w(x)$ and observable neutral anchor m^* (the margin at which $s = \frac{1}{2}$). Put $T_E := \int_D c d\mu$ (observable), and let E denote the law of all label-free observables; two targets are *evidence-identical* when they share E (equivalently (μ, m, s) , since predictions and s are fixed maps of x).

Definition 2 (Margin-monotone relative-calibration class). $\mathcal{C}_{\text{mono}}$ is the class of targets for which $a(x) = \sigma c(x)$ μ -a.e. on D for a single global orientation $\sigma \in \{\pm 1\}$; equivalently the benefit is single-crossing in the observable margin at m^* ($\text{sign } a(x) = \sigma \text{sign}(m(x) - m^*)$) with calibrated magnitude $|a(x)| = |c(x)|$. The class is *falsifiable but not verifiable*: a finite labelled probe can reject “ $a = \pm c$ ”, yet σ is never label-free identified.

Assumption 1 (General position). Within an evidence fibre the regions of D whose benefit sign is free carry comparable, not-fully- E -pinned benefit mass: $T_E \neq 0$ and no proper subcollection of free regions has an E -certifiable dominant total $\int |a| d\mu$.

Lemma 4 (Structural sign-freedom = bit complexity). *Fix E and a falsifiable class $\mathcal{C}$, and let $K(\mathcal{C}, E)$ be the number*

of disjoint μ -positive regions of D whose benefit signs are jointly free in $\mathcal{C}$ on the fibre E . Under Assumption 1, a falsifiable structural supplement of b bits pins sign Δ across the fibre iff $b \geq K(\mathcal{C}, E)$.

Proof. If $b \geq K$, declaring the orientation of each free region fixes the sign field; the rest of a being E -pinned, $\Delta = \int_D a d\mu$ then has E -computable sign. If $b < K$, some free region R is undeclared; the two targets agreeing off R and differing in sign a on R are evidence-identical and share all b declared bits, while by comparability of $\int_R |a|$ with the complement (Assumption 1) their benefits have opposite sign—no decoder of the b declared bits separates them. $\square$

Theorem 7 (Conditional resolution of Conjecture 1: a weakest one-bit class). (i) Sufficiency (*unconditional*): every target in $\mathcal{C}_{\text{mono}}$ has sign $\Delta = \sigma \text{sign}(T_E)$ with T_E observable, so E and the single orientation bit σ determine sign Δ . (ii) Necessity (*unconditional*): σ and $-\sigma$ are evidence-identical ($\text{Law}(E \mid \sigma) = \text{Law}(E \mid -\sigma)$) yet give Δ and $-\Delta$, so by *Le Cam* at least one bit is required. Thus exactly one bit certifies sign Δ on $\mathcal{C}_{\text{mono}}$, with no further assumption. (iii) Minimality (*under Assumption 1*): every falsifiable $\mathcal{C} \supsetneq \mathcal{C}_{\text{mono}}$ contains a non-margin-monotone target with $K(\mathcal{C}, E) \geq 2$, so by Lemma 4 at least two bits are required. Hence $\mathcal{C}_{\text{mono}}$ is a weakest falsifiable one-bit class—minimal in the single-crossing, calibrated-magnitude direction; it is not unique (Remark 1).

Proof. (i) $\Delta = \int_D a d\mu = \sigma \int_D c d\mu = \sigma T_E$. (ii) Sending $\sigma \mapsto -\sigma$ is the D -local label complement of Theorem 5: it fixes (μ, m, s) , hence E , while negating a and thus Δ ; the two-point bound of Theorem 2 gives minimax committal error $\frac{1}{2}$. (iii) A target outside $\mathcal{C}_{\text{mono}}$ violates sign $a = \sigma \text{sign}(m - m^*)$ on a μ -positive region, so its sign field is not one global orientation; in the swap-closed (falsifiable) fibre at least two regions are independently signable, i.e. $K \geq 2$, and Lemma 4 applies. $\square$

Proposition 2 (Explicit non-identifiability witness). For each candidate single structural bit there are two targets with identical evidence law ($\text{TV} = 0$) sharing that bit’s value but with opposite sign Δ . Take $D = R_1 \cup R_2$, $\mu(R_1) = \mu(R_2) = \frac{1}{2}$, with R_1 calibrated ($c_1=1$) and R_2 at the anchor ($c_2=0$, no observable benefit signal). For $\beta = \text{sign } a_1$: $a = (+1, -2)$ vs. $(+1, +2)$ give $\Delta = -\frac{1}{2}$ vs. $+\frac{3}{2}$. For $\beta = \text{sign } a_2$: $a = (+1, +\frac{1}{5})$ vs. $(-1, +\frac{1}{5})$ give $\Delta = +\frac{3}{5}$ vs. $-\frac{2}{5}$. No single bit certifies sign Δ , so two bits are necessary off $\mathcal{C}_{\text{mono}}$.

Remark 1 (What is unconditional, and what rests on general position). Parts (i)–(ii)—that one declared bit certifies sign Δ on $\mathcal{C}_{\text{mono}}$ —hold *unconditionally*. Only minimality (iii) uses Assumption 1: if it fails (one region’s benefit mass is E -certifiably dominant) that region’s orientation alone certifies sign Δ even outside $\mathcal{C}_{\text{mono}}$, moving the boundary. Equivalently, Assumption 1 is the requirement that the certificate be *domination-free* (minimax-sound

over the unobserved magnitudes); so we claim a *conditional* resolution of Conjecture 1 and the unconditional weakest class remains open. Even under Assumption 1, $\mathcal{C}_{\text{mono}}$ is only a weakest one-bit class: any calibrated sign-aligned field $\{a = \sigma h : h \text{ observable}\}$ is equally one-bit and is incomparable to $\mathcal{C}_{\text{mono}}$. All six checks are machine-checked by a deterministic release validator: the unconditional core (sufficiency/necessity 20000/20000, $|\Delta - \sigma T_E| = 0$ exactly), the explicit minimality witnesses of (iii), the general-position ablation, and the non-uniqueness check.

Why general position cannot simply be dropped (the precise obstruction, and exactly what is open). The minimal counterexample is two equal-mass regions $D = R_0 \cup R_1$, $\mu(R_0) = \mu(R_1) = \frac{1}{2}$, with the observable field $c = +1$ on the calibrated region R_0 and $c = 0$ on the anchor region R_1 (no observable benefit signal there, so R_1 is sign-free in any evidence fibre). Enlarge $\mathcal{C}_{\text{mono}}$ to the falsifiable class $\mathcal{C}_{\text{dom}} = \{a_{R_0} = \sigma, |a_{R_1}| \leq \rho\}$ for a declared constant $\rho < 1$. Since $\Delta = \frac{1}{2}(a_{R_0} + a_{R_1})$ and $|a_{R_1}| \leq \rho < 1 = |a_{R_0}|$, the calibrated region dominates and sign $\Delta = \sigma$ for every member, so one declared bit certifies sign Δ on all of $\mathcal{C}_{\text{dom}}$. Yet $\mathcal{C}_{\text{dom}} \supsetneq \mathcal{C}_{\text{mono}}$ (it contains non-margin-monotone targets with $a_{R_1} \neq 0$, e.g. $a = (+1, \pm\frac{1}{2})$ giving $\Delta = \frac{3}{4}, \frac{1}{4}$, both positive) and is falsifiable (a labelled probe rejects targets that are uncalibrated on R_0 or violate $|a_{R_1}| \leq \rho$). This *contradicts* minimality once an E -certifiable dominant region is permitted: dropping $a = (+1, \pm 2) \Rightarrow \Delta = +\frac{3}{2}, -\frac{1}{2}$ shows that the moment $\rho \geq 1$ the anchor region can flip sign Δ and two bits become necessary ($K=2$). The obstruction to an *unconditional* weakest class is therefore exactly this: characterising minimality across the entire lattice of dominant-region thresholds ρ (where the weakest one-bit class is not $\mathcal{C}_{\text{mono}}$ but a ρ -dependent family), which Assumption 1 sidesteps by fiat. We leave this open and do not claim it. The construction is machine-checked (`val_conj1_genpos.py`, seeds fixed): $\mathcal{C}_{\text{dom}}$ strictly contains $\mathcal{C}_{\text{mono}}$, is falsifiable (reject rate 1.0), is one-bit-certified (error rate 0 over 2×10^5 targets), and loses one-bit certifiability without the domination bound (error rate $\approx \frac{1}{4}$).

REFERENCES

- [1] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, T. Darrell. Tent: Fully test-time adaptation by entropy minimization. *ICLR*, 2021.
- [2] J. Liang, D. Hu, J. Feng. Do we really need to access the source data? Source hypothesis transfer for unsupervised domain adaptation. *ICML*, 2020.
- [3] Y. Sun, X. Wang, Z. Liu, J. Miller, A. Efros, M. Hardt. Test-time training with self-supervision for generalization under distribution shifts. *ICML*, 2020.
- [4] S. Niu, J. Wu, Y. Zhang, Y. Chen, S. Zheng, P. Zhao, M. Tan. Efficient test-time model adaptation without forgetting. *ICML*, 2022.
- [5] S. Niu, J. Wu, Y. Zhang, Z. Wen, Y. Chen, P. Zhao, M. Tan. Towards stable test-time adaptation in dynamic wild world. *ICLR*, 2023.
- [6] Q. Wang, O. Fink, L. Van Gool, D. Dai. Continual test-time domain adaptation. *CVPR*, 2022.

- [7] M. Zhang, S. Levine, C. Finn. MEMO: Test time robustness via adaptation and augmentation. *NeurIPS*, 2022.
- [8] S. Jeon, Z. Mai, H. Tian, V. Bakshi, L. Wang, J. Hou, P. Zhang, A. Chowdhury, W.-L. Chao. Continually adapt or not (CAN)? A continual learning benchmark of camera trap species classification over time. *NeurIPS Workshop (Imageomics)*, 2025. OpenReview: [jEItIzAgHs](#).
- [9] M. Schirmer, M. Jazbec, C. A. Naeseth, E. Nalisnick. Monitoring risks in test-time adaptation. *NeurIPS*, 2025. arXiv:2507.08721.
- [10] A. Podkopaev, A. Ramdas. Tracking the risk of a deployed model and detecting harmful distribution shifts. *ICLR*, 2022. arXiv:2110.06177.
- [11] Y. Bar, S. Shaer, Y. Romano. Protected test-time adaptation via online entropy matching: A betting approach. *NeurIPS*, 2024. arXiv:2408.07511.
- [12] A. Sreeram, M. N. Huh, J. K. Wong, et al. Tempora: Characterising the time-contingent utility of online test-time adaptation. arXiv:2602.06136, 2026.
- [13] S. Garg, S. Balakrishnan, Z. Lipton, B. Neyshabur, H. Sedghi. Leveraging unlabeled data to predict out-of-distribution performance. *ICLR*, 2022. arXiv:2201.04234.
- [14] J. Miller, R. Taori, A. Raghunathan, S. Sagawa, P. W. Koh, V. Shankar, P. Liang, Y. Carmon, L. Schmidt. Accuracy on the line: On the strong correlation between out-of-distribution and in-distribution generalization. *ICML*, 2021. arXiv:2107.04649.
- [15] C. Baek, Y. Jiang, A. Raghunathan, J. Z. Kolter. Agreement-on-the-line: Predicting the performance of neural networks under distribution shift. *NeurIPS*, 2022. arXiv:2206.13089.
- [16] E. Kim, M. Sun, C. Baek, A. Raghunathan, J. Z. Kolter. Test-time adaptation induces stronger accuracy and agreement-on-the-line. *NeurIPS*, 2024. arXiv:2310.04941.
- [17] T. Lee, S. Chottananurak, T. Gong, S.-J. Lee. AETTA: Label-free accuracy estimation for test-time adaptation. *CVPR*, 2024. arXiv:2404.01351.
- [18] Y. Jiang, V. Nagarajan, C. Baek, J. Z. Kolter. Assessing generalization of SGD via disagreement. *ICLR*, 2022.
- [19] Y. Lu, Z. Wang, R. Zhai, S. Kolouri, J. Campbell, K. Sycara. Characterizing out-of-distribution error via optimal transport. *NeurIPS*, 2023.
- [20] J. Steinhardt, P. Liang. Unsupervised risk estimation using only conditional independence structure. *NeurIPS*, 2016. arXiv:1606.05313.
- [21] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, J. Wortman Vaughan. A theory of learning from different domains. *Machine Learning* 79(1-2):151–175, 2010.
- [22] S. Ben-David, T. Lu, T. Luu, D. Pál. Impossibility theorems for domain adaptation. *AISTATS*, 2010.
- [23] Z. Lipton, Y.-X. Wang, A. Smola. Detecting and correcting for label shift with black box predictors. *ICML*, 2018.
- [24] E. Rosenfeld, S. Garg. (Almost) provable error bounds under distribution shift via disagreement discrepancy. *NeurIPS*, 2023. arXiv:2306.00312.
- [25] F. Parisi, F. Strino, B. Nadler, Y. Kluger. Ranking and combining multiple predictors without labeled data. *PNAS* 111(4):1253–1258, 2014. arXiv:1303.3257.
- [26] A. Jaffe, B. Nadler, Y. Kluger. Estimating the accuracies of multiple classifiers without labeled data. *AISTATS*, 2015. arXiv:1407.7644.
- [27] A. Jaffe, E. Fetaya, B. Nadler, T. Jiang, Y. Kluger. Unsupervised ensemble learning with dependent classifiers. *AISTATS*, 2016. arXiv:1510.05830.
- [28] E. A. Platanios, A. Blum, T. Mitchell. Estimating accuracy from unlabeled data. *UAI*, 2014.
- [29] S. Ibrahim, X. Fu, N. Sidiropoulos. Crowdsourcing via pairwise co-occurrences: Identifiability and algorithms. *NeurIPS*, 2019. arXiv:1909.12325.
- [30] A. Corrada-Emmanuel. The logic of NTQR evaluations of noisy AI agents: Complete postulates and logically consistent error correlations. *NeurIPS*, 2024. arXiv:2312.05392; see also arXiv:2409.11052, arXiv:2412.16238.
- [31] A. P. Dawid, A. M. Skene. Maximum likelihood estimation of observer error-rates using the EM algorithm. *JRSS-C* 28(1):20–28, 1979.
- [32] L. Le Cam. *Asymptotic Methods in Statistical Decision Theory*. Springer, 1986.
- [33] C. Manski. Nonparametric bounds on treatment effects. *Amer. Econ. Rev. P&P* 80(2):319–323, 1990.
- [34] Y. Geifman, R. El-Yaniv. Selective classification for deep neural networks. *NeurIPS*, 2017.
- [35] V. Vovk, A. Gammerman, G. Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.
- [36] A. Angelopoulos, S. Bates. Conformal prediction: A gentle introduction. *Found. Trends Mach. Learn.*, 2023.
- [37] G. Shafer. Testing by betting: A strategy for statistical and scientific communication. *JRSS-A*, 2021.
- [38] A. Ramdas, P. Grünwald, V. Vovk, G. Shafer. Game-theoretic statistics and safe anytime-valid inference. *Statist. Sci.*, 2023.
- [39] V. Vovk, I. Petej, I. Nouretdinov, E. Ahlberg, L. Carlsson, A. Gammerman. Retrain or not retrain: Conformal test martingales for change-point detection. *COPA*, 2021. arXiv:2102.10439.
- [40] S. R. Howard, A. Ramdas, J. McAuliffe, J. Sekhon. Time-uniform, nonparametric, nonasymptotic confidence sequences. *Annals of Statistics* 49(2):1055–1080, 2021.
- [41] A. Maurer, M. Pontil. Empirical Bernstein bounds and sample variance penalization. *COLT*, 2009.
- [42] W. Hoeffding. Probability inequalities for sums of bounded random variables. *JASA* 58(301):13–30, 1963.
- [43] A. Klivans, K. Stavropoulos, A. Vasilyan. Testable learning with distribution shift. arXiv:2311.15142, 2023.
- [44] A. Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.
- [45] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, L. Fei-Fei. ImageNet: A large-scale hierarchical image database. *CVPR*, 2009.
- [46] D. Hendrycks, T. Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. *ICLR*, 2019.
- [47] D. Hendrycks, S. Basart, N. Mu, S. Kadavath, F. Wang, E. Dorundo, R. Desai, W. Zhu, R. Parajuli, M. Guo, D. Song, J. Steinhardt. The many faces of robustness: A critical analysis of out-of-distribution generalization. *ICCV*, 2021.
- [48] P. W. Koh, S. Sagawa, H. Marklund, S. M. Xie, M. Zhang, A. Balsubramani, W. Hu, M. Yasunaga, et al. WILDS: A benchmark of in-the-wild distribution shifts. *ICML*, 2021.
- [49] P. Bándi, O. Geessink, Q. Manson, M. van Dijk, M. Balkenhol, M. Hermesen, et al. From detection of individual metastases to classification of lymph node status at the patient level: The CAMELYON17 challenge. *IEEE Trans. Med. Imaging* 38(2):550–560, 2019.
- [50] J. Taylor, S. L. Earnshaw, B. M. Mabey, M. Victors, M. N. Annis, D. Y. Torigian, et al. The RXX1 dataset and challenge: Unsupervised classification of cellular morphological changes. *arXiv:1907.04758*, 2019.
- [51] H. Venkateswara, J. Eusebio, S. Chakraborty, S. Panchanathan. Deep hashing network for unsupervised domain adaptation. *CVPR*, 2017. (Office-Home dataset release).
- [52] I. Gulrajani, D. Lopez-Paz. In search of lost domain generalization. *ICLR*, 2021.
- [53] F. Croce, M. Andriushchenko, V. Schwag, E. Debenedetti, N. Flammarion, M. Chiang, P. Mittal, M. Hein. RobustBench: A standardized adversarial robustness benchmark. *NeurIPS Datasets and Benchmarks*, 2021.
- [54] B. Recht, R. Roelofs, L. Schmidt, V. Shankar. Do CIFAR-10 classifiers generalize to CIFAR-10? *ICML*, 2019.
- [55] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, et al. PyTorch: An imperative style, high-performance deep learning library. *NeurIPS*, 2019.

- [56] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, et al. Scikit-learn: Machine learning in Python. *JMLR* 12:2825–2830, 2011.
- [57] B. Schölkopf, D. Janzing, J. Peters, E. Sgouritsa, K. Zhang, J. Mooij. On causal and anticausal learning. *ICML*, 2012.
- [58] X. Wu, M. Gong, J. H. Manton, U. Aickelin, J. Zhu. On causality in domain adaptation and semi-supervised learning: an information-theoretic analysis for parametric models. *JMLR* 25(261):1–57, 2024.